

Problem

Prove that the g parameters can be obtained from the z parameters as

$$\begin{aligned} g_{11} &= \frac{1}{z_{11}}, & g_{12} &= -\frac{z_{12}}{z_{11}} \\ g_{21} &= \frac{z_{21}}{z_{11}}, & g_{22} &= \frac{\Delta_z}{z_{11}} \end{aligned}$$

Step-by-step solution

Step 1 of 2

The Z - parameter equation are

$$V_1 = Z_{11}I_1 + Z_{12}I_2 \quad -(i)$$

$$V_2 = Z_{21}I_1 + Z_{22}I_2 \quad -(ii)$$

Step 2 of 2

From equation (i)

$$\begin{aligned} V_2 &= Z_{21}I_1 + Z_{22}I_2 \\ &= Z_{21} \left[\frac{V_1}{Z_{11}} - \left(\frac{Z_{12}}{Z_{11}} \right) I_2 \right] + Z_{22}I_2 \\ &= \left(\frac{Z_{21}}{Z_{11}} \right) V_1 + \left(\frac{Z_{11}Z_{22} - Z_{12}Z_{21}}{Z_{11}} \right) I_2 \\ &= \left(\frac{Z_{21}}{Z_{11}} \right) V_1 + \left(\frac{\Delta Z}{Z_{11}} \right) I_2 \\ &\equiv g_{21}V_1 + g_{22}I_2 \end{aligned}$$

So we have

$$g_{11} = \frac{1}{Z_{11}} \quad g_{12} = -\frac{Z_{12}}{Z_{11}} \quad g_{21} = \frac{Z_{21}}{Z_{11}} \quad g_{22} = \frac{\Delta Z}{Z_{11}}$$